



Final Exam Sem 3 2019 2020

Statistics and Probability (Universiti Teknologi Malaysia)



messages.pdf_cover_qr_code_label

QUESTION 1 (6 marks)

The following table shows the reading time (to the nearest minute) spent during a particular day (for a group of school children):

Time (X)	Number of children
$10 \leq X < 15$	8
$15 \leq X < 20$	15
$20 \leq X < 25$	K
$25 \leq X < 30$	18
$30 \leq X < 35$	12
$35 \leq X < 40$	7
$40 \leq X < 45$	5

If the mean of the reading time for this group of school children is $\frac{224}{9}$ minutes;

- i. show that $K = 25$ (3 marks)
- ii. compute the standard deviation for the above data. (3 marks)

QUESTION 2 (6 MARKS)

How many different ways can the letters of the word POPULAR be arranged such that

- i. begin and end with P (1 mark)
- ii. have the two P 's separated (2 marks)
- iii. have the vowels together. (3 marks)

QUESTION 3 (12 MARKS)

- a) For a certain population of students, the percentage of passing and failing an examination is as shown below. Of all students taking the exam 36% are in the sponsored - pass category, 24% are in the sponsored - fail category, and so forth. A student is selected randomly from the population. Let L be the event that the student passes the exam and S be the event that the student is sponsored.

Outcome	Sponsored (S)	Not sponsored (S')	Total
Pass (L)	36	24	60
Fail (L')	24	16	40
Total	60	40	100

- i. Determine the probability that the student is either sponsored or fail the exam. (2 marks)
 - ii. Find the probability that the student passes the exam given that he is not sponsored. (3 marks)
 - iii. Are the event L and S independent? State your reason. (2 marks)
- b) A diagnostic test for a certain disease is said to be 95% accurate such that if a person has the disease, the test will show positive with probability 0.95, and if he does not have the disease, the test will show negative with probability 0.95. Only 2% of the population has the disease. A person is tested at random from the population.
- i. Determine the probability that he is tested positive. (2 marks)
 - ii. Determine the probability that he does not have the disease given that he is tested positive. (3 marks)

QUESTION 4 (11 MARKS)

- a) The probability distribution of a discrete random variable X is given below.

x	1	2	3
$P(X = x)$	a	b	c

Given that $E(X) = 1.7$ and $Var(X) = 0.81$, calculate the values of a , b , and c .

(5 marks)

b) The continuous random variable X has the following probability density function.

$$f(x) = \begin{cases} x + 1 ; & 0 \leq x < 1 \\ 1 - x ; & 1 \leq x < 2 \\ 0 ; & \text{otherwise} \end{cases}$$

i. Find the cumulative distribution function of X , $F(X)$. (4 marks)

ii. Find $P\left(\frac{1}{2} < X \leq \frac{3}{2}\right)$. (2 marks)

QUESTION 5 (12 MARKS)

a) A manufacturer produces bottles of mineral water. The mineral water is sold in boxes containing 25 bottles of mineral water each. The probability that a bottle of mineral water from a box is defective is 0.1. A box is selected at random.

i. Find the probability that no bottle of mineral water is defective. (2 marks)

ii. Find the probability that between three and seven bottles are defective. (2 marks)

iii. A company purchases 10 boxes containing bottles of mineral water from the manufacturer. What is the probability that at least one of the boxes contains no defective bottles of mineral water? (3 marks)

b) A manufacturer of an electronics product suspected that 1% of their products would fail during the warranty period. A sample of 400 independent units is checked for their performance. Using a suitable approximation

i. find the probability that not more than one unit fails during the warranty period (3 marks)

ii. compute the probability that more than ten units fail during the warranty period. (2 marks)

QUESTION 6 (19 MARKS)

- a) The number of claims received by an insurance company in a month follows a Poisson distribution with mean m . If the probability that at least one claim has been received in a month is 0.97,
- compute m (round the answer to one decimal point) (3 marks)
 - compute the probability that at least eight claims are received in a month (2 marks)
 - compute the probability between 70 and 80 claims are received in 2 years using a suitable approximation. (4 marks)
- b) Vehicles pass through a junction on a busy road at an average rate of 300 per hour.
- Find the probability that none of the vehicles passes through the junction in a given minute. (2 marks)
 - What is the expected number of vehicles passing through the junction in five minutes? (1 mark)
 - Compute the probability that exactly 20 vehicles pass through the junction in five minutes. (2 marks)
- c) Flaws that occur in the interior of plastic used for automobiles follow a Poisson distribution with a mean 0.02 flaw per panel.
- Find the probability that there is no flaw if 20 panels are inspected. (2 marks)
 - If 20 panels are inspected, find the probability that at most two panels have at least one flaw. (3 marks)

QUESTION 7 (19 MARKS)

- a) The download time to be completed of a resource web page is normally distributed with a mean of 6.5 seconds and a standard deviation of 2.3 seconds.
- Find the probability that download time will take less than 5 seconds. (3 marks)

- ii. What is the probability that the download time will be between 4 and 10 seconds? (3 marks)
 - iii. How much time does it take for 35% of the downloads to be completed? (3 marks)
- b) In a company, the wages of a certain grade of staff are normally distributed with a standard deviation of RM 400. If 20.05% of staff earn less than RM 300 a week, find the average wage, μ . (4 marks)
- c) A road survey has shown that the speeds of cars along a road are normally distributed with a mean speed of 97km/h and a standard deviation of 14km/h. Compute the interquartile range for this distribution. (6 marks)

QUESTION 8 (15 MARKS)

- a) Data recorded from a high school show that 80% of the student is right-handed.
- i. A random sample of 10 students is selected. Compute the probability that more than three students are right-handed. (3 marks)
 - ii. A random sample of 150 students is selected. Using a suitable approximation to determine the probability that at least 125 of them are right-handed. (4 marks)
- b) A hardware store sells plank wood with defects. The defects occur at random on the wood with a constant rate of 5 per 10-centimeter length.
- i. Chong buys a plank of length 1 meter. Find the probability that Chong's plank contains at most 30 defects in 1 meter. (4 marks)
 - ii. What is the probability that the defects are between 190 and 230 when Chong buys 4-meter plank wood? (4 marks)

LIST OF FORMULA

Descriptive Statistics and Probability

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

$$\sigma^2 = \frac{1}{N} \left(\sum_{i=1}^N x_i^2 - \frac{\left(\sum_{i=1}^N x_i \right)^2}{N} \right)$$

$$\sigma = \sqrt{\sigma^2}$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$s^2 = \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i \right)^2}{n} \right)$$

$$s = \sqrt{s^2}$$

$$\mu = \frac{\sum_{i=1}^N f_i x_i}{\sum_{i=1}^N f_i}$$

$$\sigma^2 = \frac{1}{\sum_{i=1}^N f_i} \left(\sum_{i=1}^N f_i x_i^2 - \frac{\left(\sum_{i=1}^N f_i x_i \right)^2}{\sum_{i=1}^N f_i} \right)$$

$$\sigma = \sqrt{\sigma^2}$$

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}$$

$$s^2 = \frac{1}{\sum_{i=1}^n f_i - 1} \left(\sum_{i=1}^n f_i x_i^2 - \frac{\left(\sum_{i=1}^n f_i x_i \right)^2}{\sum_{i=1}^n f_i} \right)$$

$$s = \sqrt{s^2}$$

$$\text{Median : } L + \left(\frac{\frac{n}{2} - F}{f} \right) \times c$$

$$P_i = L + \left(\frac{\frac{i}{100}(n) - F}{f} \right) \times c$$

$$\text{Mode: } L + \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right) \times c$$

$$Q_i = L + \left(\frac{\frac{i}{4}(n) - F}{f} \right) \times c$$

$$D_i = L + \left(\frac{\frac{i}{10}(n) - F}{f} \right) \times c$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, P(B) > 0.$$

$${}^nC_r = \binom{n}{r} = \frac{n!}{(n-r)!r!}$$

$${}^nP_r = \binom{n}{r} = \frac{n!}{(n-r)!}$$

Random Variable

$$E(X) = \sum xP(X = x)$$

$$E(X) = \int xf(x)dx$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Probability distributions

Distribution	Probability functions	Mean	Variance
Binomial	$P(X = x) = \binom{n}{x} \pi^x (1-\pi)^{n-x}$ $x = 0, 1, \dots, n, \quad 0 \leq \pi \leq 1$	$n\pi$	$n\pi(1-\pi)$
Poisson	$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$ $x = 0, 1, 2, \dots, \quad \lambda > 0$	λ	λ
Normal	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ $-\infty < x < \infty, \quad -\infty < \mu < \infty, \quad \sigma > 0.$	μ	σ^2
Standard Normal	$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$		